

1 NF

mysql> Select * from ProjectDetails;

ProjectID	ClientID	ClientName	ClientEmail	FreelancerID	FreelancerName	FreelancerSkill	ProjectTitle	Category	Budget	PaymentID	PaymentDate	Amount
100	1	Amit Sharma	amit@gmail.com	201	Rahul Sharma	Web Development	E-Commerce Website	Web Development	50000.00	301	2024-06-01	35000.00
100	1	Amit Sharma	amit@gmail.com	201	Rahul Sharma	Web Development	E-Commerce Website	Web Development	50000.00	302	2024-06-28	20000.00
102	2	Neha Kapoor	nehag@gmail.com	202	Priya Singh	Graphic Design	Logo Design	Design	12000.00	303	2024-06-05	12000.00
103	1	Amit Sharma	amit@gmail.com	203	Aarav Sharma	Content Writing	Blog Writing	Writing	8000.00	304	2024-06-12	6000.00
104	3	Rohan Gupta	rohan@gmail.com	201	Rahul Sharma	Web Development	Portfolio Website	Web Development	18000.00	305	2024-06-18	9000.00
104	3	Rohan Gupta	rohan@gmail.com	201	Rahul Sharma	Web Development	Portfolio Website	Web Development	18000.00	306	2024-07-01	9000.00
105	4	Swaha Arora	swahag@gmail.com	204	Karen Malhotra	Mobile App Development	Food Delivery App	Mobile Development	70000.00	307	2024-06-15	30000.00
106	4	Swaha Arora	swahag@gmail.com	204	Karen Malhotra	Mobile App Development	Food Delivery App	Mobile Development	70000.00	308	2024-07-05	40000.00
106	2	Neha Kapoor	nehag@gmail.com	202	Priya Singh	Graphic Design	Business Card Design	Design	7000.00	309	2024-07-02	7000.00
107	5	Aditya Jain	adityaj@gmail.com	205	Meera Joshi	Video Editing	YouTube Video Editing	Editing	25000.00	310	2024-07-04	20000.00

10 rows in set (0.00 sec)

Primary Key
(ProjectID, PaymentID)

Candidate Key
(ProjectID, PaymentID)

Redundancy

```
mysql> SELECT ClientID, ClientName, COUNT(*) AS Total
-> FROM ProjectDetails
-> GROUP BY ClientID, ClientName
[ -> HAVING COUNT(*) > 1;
```

ClientID	ClientName	Total
1	Amit Sharma	3
2	Neha Kapoor	2
3	Rohan Gupta	2
4	Sneha Arora	2

4 rows in set (0.00 sec)

```
mysql> SELECT FreelancerID, FreelancerName, COUNT(*) AS Total
-> FROM ProjectDetails
-> GROUP BY FreelancerID, FreelancerName
[ -> HAVING COUNT(*) > 1;
```

FreelancerID	FreelancerName	Total
201	Rahul Verma	4
202	Priya Singh	2
204	Karan Malhotra	2

3 rows in set (0.01 sec)

```
mysql> SELECT ProjectID, ProjectTitle, COUNT(*) AS Total
-> FROM ProjectDetails
-> GROUP BY ProjectID, ProjectTitle
[ -> HAVING COUNT(*) > 1;
```

ProjectID	ProjectTitle	Total
101	E-Commerce Website	2
104	Portfolio Website	2
105	Food Delivery App	2

3 rows in set (0.00 sec)

Functional Dependency

```
mysql> SELECT ProjectID,
-> COUNT(DISTINCT ProjectTitle) AS Titles
-> FROM ProjectDetails
-> GROUP BY ProjectID;
```

ProjectID	Titles
101	1
102	1
103	1
104	1
105	1
106	1
107	1

7 rows in set (0.00 sec)

```
mysql> SELECT ClientID,
-> COUNT(DISTINCT ClientName)
-> FROM ProjectDetails
-> GROUP BY ClientID;
```

ClientID	COUNT(DISTINCT ClientName)
1	1
2	1
3	1
4	1
5	1

5 rows in set (0.00 sec)

```
mysql> SELECT FreelancerID,
-> COUNT(DISTINCT FreelancerSkill)
-> FROM ProjectDetails
-> GROUP BY FreelancerID;
```

FreelancerID	COUNT(DISTINCT FreelancerSkill)
201	1
202	1
203	1
204	1
205	1

5 rows in set (0.00 sec)

```
mysql> SELECT PaymentID,
-> COUNT(DISTINCT PaymentDate)
-> FROM ProjectDetails
-> GROUP BY PaymentID;
```

PaymentID	COUNT(DISTINCT PaymentDate)
301	1
302	1
303	1
304	1
305	1
306	1
307	1
308	1
309	1
310	1

10 rows in set (0.00 sec)

2 NF

```
mysql> Select * from Projects2NF;
```

ProjectID	ProjectTitle	Category	Budget	ClientID	ClientName	ClientEmail	FreelancerID	FreelancerName	FreelancerSkill
301	E-Commerce Website	Web Development	50000.00	1	Anil Sharma	anilsgmail.com	201	Rahul Verma	Web Development
302	Logo Design	Design	12000.00	2	Roha Kapoor	roha@gmail.com	202	Priya Singh	Graphic Design
303	Blog Writing	Writing	5000.00	1	Anil Sharma	anilsgmail.com	203	Arjun Verma	Content Writing
304	Portfolio Website	Web Development	30000.00	3	Rohan Gupta	rohan@gmail.com	201	Rahul Verma	Web Development
305	Food Delivery App	Mobile Development	75000.00	4	Shruti Kishor	shrutik@gmail.com	204	Karan Mahotra	Mobile App Development
306	Business Card Design	Design	7000.00	2	Roha Kapoor	roha@gmail.com	202	Priya Singh	Graphic Design
307	YouTube Video Editing	Editing	35000.00	5	Aditya Jain	adityaj@gmail.com	205	Meera Joshi	Video Editing

```
7 rows in set (0.00 sec)
```

```
[mysql> Select * from Payments2NF;
```

PaymentID	ProjectID	PaymentDate	Amount
301	101	2026-06-01	25000.00
302	101	2026-06-20	25000.00
303	102	2026-06-05	12000.00
304	103	2026-06-12	8000.00
305	104	2026-06-18	9000.00
306	104	2026-07-01	9000.00
307	105	2026-06-15	35000.00
308	105	2026-07-05	40000.00
309	106	2026-07-02	7000.00
310	107	2026-07-04	15000.00

```
10 rows in set (0.00 sec)
```

3 NF

```
mysql> SELECT * FROM Clients;
```

ClientID	ClientName	ClientEmail
1	Amit Sharma	amit@gmail.com
2	Neha Kapoor	neha@gmail.com
3	Rohan Gupta	rohan@gmail.com
4	Sneha Arora	sneha@gmail.com
5	Aditya Jain	aditya@gmail.com

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM Freelancers;
```

FreelancerID	FreelancerName	FreelancerSkill
201	Rahul Verma	Web Development
202	Priya Singh	Graphic Design
203	Arjun Mehta	Content Writing
204	Karan Malhotra	Mobile App Development
205	Meera Joshi	Video Editing

5 rows in set (0.00 sec)

```
mysql> SELECT * FROM Projects;
```

ProjectID	ProjectTitle	Category	Budget	ClientID	FreelancerID
101	E-Commerce Website	Web Development	50000.00	1	201
102	Logo Design	Design	12000.00	2	202
103	Blog Writing	Writing	8000.00	1	203
104	Portfolio Website	Web Development	18000.00	3	201
105	Food Delivery App	Mobile Development	75000.00	4	204
106	Business Card Design	Design	7000.00	2	202
107	YouTube Video Editing	Editing	15000.00	5	205

7 rows in set (0.00 sec)

```
mysql> SELECT * FROM Payments;
```

PaymentID	ProjectID	PaymentDate	Amount
301	101	2026-06-01	25000.00
302	101	2026-06-20	25000.00
303	102	2026-06-05	12000.00
304	103	2026-06-12	8000.00
305	104	2026-06-18	9000.00
306	104	2026-07-01	9000.00
307	105	2026-06-15	35000.00
308	105	2026-07-05	40000.00
309	106	2026-07-02	7000.00
310	107	2026-07-04	15000.00

10 rows in set (0.00 sec)

Original (Unnormalized) Database Design

```
mysql> SELECT * FROM ProjectDetails;
```

ProjectID	ClientID	ClientName	ClientEmail	FreelancerID	FreelancerName	FreelancerSkill	ProjectTitle	Category	Budget	PaymentID	PaymentDate	Amount
101	1	Amit Sharma	amit@gmail.com	201	Rahul Verma	Web Development	E-Commerce Website	Web Development	50000.00	301	2026-06-01	25000.00
101	1	Amit Sharma	amit@gmail.com	201	Rahul Verma	Web Development	E-Commerce Website	Web Development	50000.00	302	2026-06-20	25000.00
102	2	Neha Kapoor	neha@gmail.com	202	Priya Singh	Graphic Design	Logo Design	Design	12000.00	303	2026-06-05	12000.00
103	1	Amit Sharma	amit@gmail.com	203	Arjun Mehta	Content Writing	Blog Writing	Writing	8000.00	304	2026-06-12	8000.00
104	3	Rohan Gupta	rohan@gmail.com	201	Rahul Verma	Web Development	Portfolio Website	Web Development	18000.00	305	2026-06-18	9000.00
104	3	Rohan Gupta	rohan@gmail.com	201	Rahul Verma	Web Development	Portfolio Website	Web Development	18000.00	306	2026-07-01	9000.00
105	4	Sneha Arora	sneha@gmail.com	204	Karan Malhotra	Mobile App Development	Food Delivery App	Mobile Development	75000.00	307	2026-06-15	35000.00
105	4	Sneha Arora	sneha@gmail.com	204	Karan Malhotra	Mobile App Development	Food Delivery App	Mobile Development	75000.00	308	2026-07-05	40000.00
106	2	Neha Kapoor	neha@gmail.com	202	Priya Singh	Graphic Design	Business Card Design	Design	7000.00	309	2026-07-02	7000.00
107	5	Aditya Jain	aditya@gmail.com	205	Meera Joshi	Video Editing	YouTube Video Editing	Editing	15000.00	310	2026-07-04	15000.00

20 rows in set (0.00 sec)

Normalized (3NF) Database Design


```
mysql> DESC Clients;
```

Field	Type	Null	Key	Default	Extra
ClientID	int	NO	PRI	NULL	
ClientName	varchar(100)	YES		NULL	
ClientEmail	varchar(100)	YES		NULL	

```
3 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Clients;
```

ClientID	ClientName	ClientEmail
1	Amit Sharma	amit@gmail.com
2	Neha Kapoor	neha@gmail.com
3	Rohan Gupta	rohan@gmail.com
4	Sneha Arora	sneha@gmail.com
5	Aditya Jain	aditya@gmail.com

```
5 rows in set (0.00 sec)
```

```
mysql> DESC Freelancers;
```

Field	Type	Null	Key	Default	Extra
FreelancerID	int	NO	PRI	NULL	
FreelancerName	varchar(100)	YES		NULL	
FreelancerSkill	varchar(100)	YES		NULL	

```
3 rows in set (0.01 sec)
```

```
mysql> SELECT * FROM Freelancers;
```

FreelancerID	FreelancerName	FreelancerSkill
201	Rahul Verma	Web Development
202	Priya Singh	Graphic Design
203	Arjun Mehta	Content Writing
204	Karan Malhotra	Mobile App Development
205	Meera Joshi	Video Editing

```
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Projects;
```

ProjectID	ProjectTitle	Category	Budget	ClientID	FreelancerID
101	E-Commerce Website	Web Development	50000.00	1	201
102	Logo Design	Design	12000.00	2	202
103	Blog Writing	Writing	8000.00	1	203
104	Portfolio Website	Web Development	18000.00	3	201
105	Food Delivery App	Mobile Development	75000.00	4	204
106	Business Card Design	Design	7000.00	2	202
107	YouTube Video Editing	Editing	15000.00	5	205

7 rows in set (0.00 sec)

```
mysql> DESC Payments;
```

Field	Type	Null	Key	Default	Extra
PaymentID	int	NO	PRI	NULL	
ProjectID	int	YES		NULL	
PaymentDate	date	YES		NULL	
Amount	decimal(10,2)	YES		NULL	

4 rows in set (0.00 sec)

```
mysql> SELECT * FROM Payments;
```

PaymentID	ProjectID	PaymentDate	Amount
301	101	2026-06-01	25000.00
302	101	2026-06-20	25000.00
303	102	2026-06-05	12000.00
304	103	2026-06-12	8000.00
305	104	2026-06-18	9000.00
306	104	2026-07-01	9000.00
307	105	2026-06-15	35000.00
308	105	2026-07-05	40000.00
309	106	2026-07-02	7000.00
310	107	2026-07-04	15000.00

10 rows in set (0.00 sec)

(Before Normalization)

Insert Operation

```
mysql> INSERT INTO ProjectDetails
-> VALUES
-> (200,4,'Rash Mehta','rash@gmail.com',
-> 204,'Aisha Khan','UI/UX Design',
-> 'Landing Page Design','Design',
-> 10000.00,'2026-07-05',10000.00);
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM ProjectDetails
-> WHERE ProjectID=200;
```

ProjectID	ClientID	ClientName	ClientEmail	FreelancerID	FreelancerName	FreelancerSkill	ProjectTitle	Category	Budget	PaymentID	PaymentDate	Amount
200	4	Rash Mehta	rash@gmail.com	204	Aisha Khan	UI/UX Design	Landing Page Design	Design	10000.00	311	2026-07-05	10000.00

1 row in set (0.00 sec)

Update Operation

```
mysql> UPDATE ProjectDetails
    -> SET FreelancerSkill='Full Stack Development'
    -> WHERE FreelancerID=201;
Query OK, 4 rows affected (0.01 sec)
Rows matched: 4  Changed: 4  Warnings: 0

mysql> SELECT FreelancerID,FreelancerName,FreelancerSkill
    -> FROM ProjectDetails
    -> WHERE FreelancerID=201;
```

FreelancerID	FreelancerName	FreelancerSkill
201	Rahul Verma	Full Stack Development
201	Rahul Verma	Full Stack Development
201	Rahul Verma	Full Stack Development
201	Rahul Verma	Full Stack Development

4 rows in set (0.00 sec)

Delete Operation

```
mysql> DELETE FROM ProjectDetails
    -> WHERE PaymentID=310;
Query OK, 1 row affected (0.00 sec)

mysql> SELECT *
    -> FROM ProjectDetails
    -> WHERE ProjectID=107;
Empty set (0.00 sec)
```

Retrieval using JOIN queries

```
mysql> SELECT ProjectTitle,
    -> ClientName,
    -> FreelancerName,
    -> Amount
    -> FROM ProjectDetails;
```

ProjectTitle	ClientName	FreelancerName	Amount
E-Commerce Website	Amit Sharma	Rahul Verma	25000.00
E-Commerce Website	Amit Sharma	Rahul Verma	25000.00
Logo Design	Neha Kapoor	Priya Singh	12000.00
Blog Writing	Amit Sharma	Arjun Mehta	8000.00
Portfolio Website	Rohan Gupta	Rahul Verma	9000.00
Portfolio Website	Rohan Gupta	Rahul Verma	9000.00
Food Delivery App	Sneha Arora	Karan Malhotra	35000.00
Food Delivery App	Sneha Arora	Karan Malhotra	40000.00
Business Card Design	Neha Kapoor	Priya Singh	7000.00
Landing Page Design	Harsh Mehta	Aisha Khan	10000.00

10 rows in set (0.00 sec)

After Normalization

Insert Operation


```

mysql> INSERT INTO Clients
-> VALUES
-> (6,'Harsh Mehta','harsh@gmail.com');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Freelancers
-> VALUES
-> (206,'Aisha Khan','UI/UX Design');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Projects
-> VALUES
-> (108,'Landing Page Design','Design',10000,6,206);
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> INSERT INTO Payments
-> VALUES
-> (311,108,'2026-07-08',10000);
Query OK, 1 row affected (0.00 sec)

mysql> SELECT * FROM Clients WHERE ClientID=6;
+-----+-----+-----+
| ClientID | ClientName | ClientEmail |
+-----+-----+-----+
| 6 | Harsh Mehta | harsh@gmail.com |
+-----+-----+-----+
1 row in set (0.00 sec)

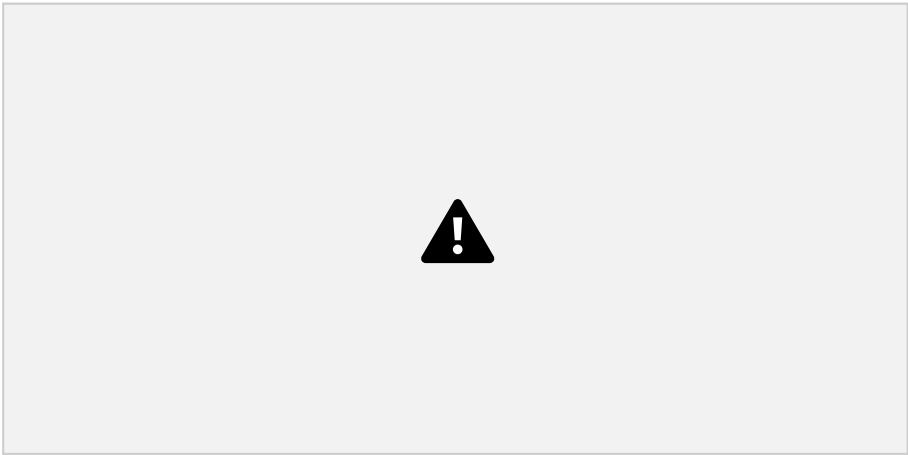
mysql>
mysql> SELECT * FROM Freelancers WHERE FreelancerID=206;
+-----+-----+-----+
| FreelancerID | FreelancerName | FreelancerSkill |
+-----+-----+-----+
| 206 | Aisha Khan | UI/UX Design |
+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
mysql> SELECT * FROM Projects WHERE ProjectID=108;
+-----+-----+-----+-----+-----+-----+
| ProjectID | ProjectTitle | Category | Budget | ClientID | FreelancerID |
+-----+-----+-----+-----+-----+-----+
| 108 | Landing Page Design | Design | 10000.00 | 6 | 206 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

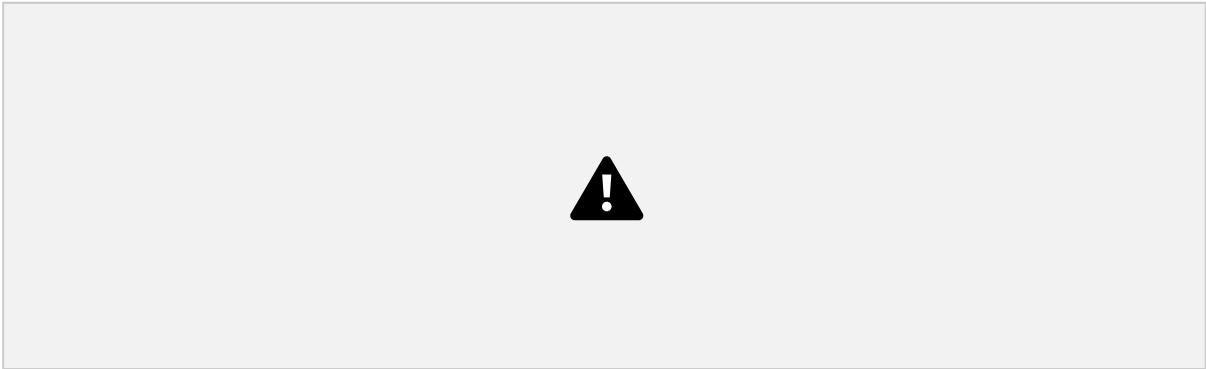
mysql>
mysql> SELECT * FROM Payments WHERE PaymentID=311;
+-----+-----+-----+-----+
| PaymentID | ProjectID | PaymentDate | Amount |
+-----+-----+-----+-----+
| 311 | 108 | 2026-07-08 | 10000.00 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

```

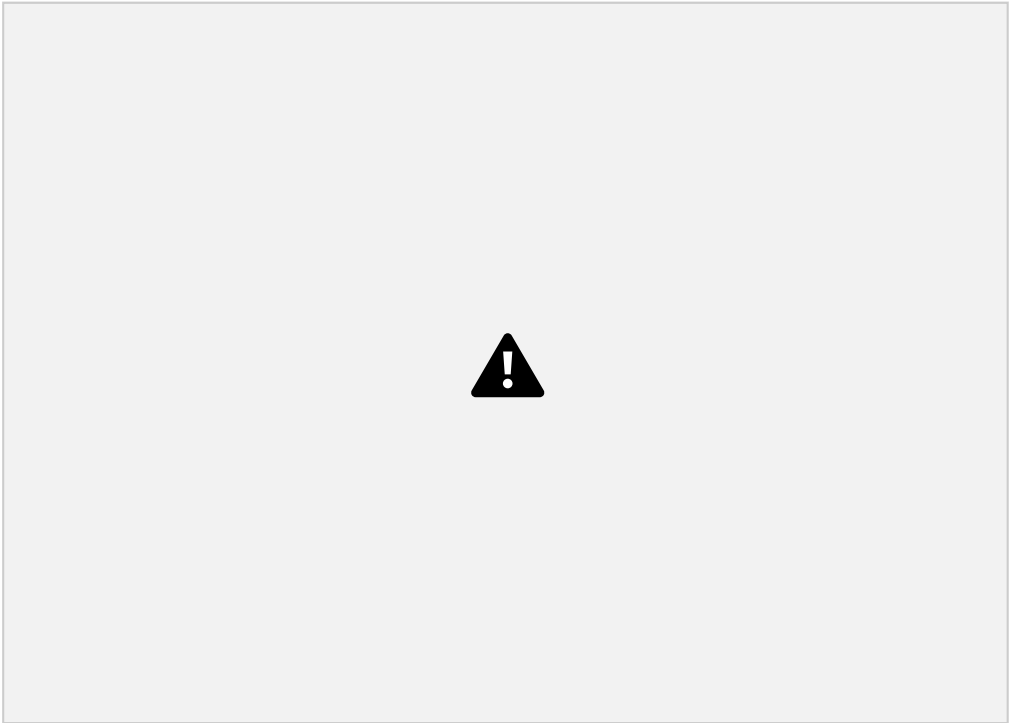
Update Operation



Delete Operation



Retrieval using JOIN queries



Parameter Before Normalization (Original Table)

After Normalization (3NF)

1 (ProjectDetails) 4 (Clients, Freelancers,
Projects, Payments)

Number of tables

project details are repeated)

Data

redundancy

High (duplicate records
consume more space)

Low (no unnecessary
duplication)

Storage

usage

Present (same data must be
updated in multiple rows)

Eliminated (update only one
record)

Update

anomaly

Present (cannot insert some
information independently)

Insert

anomaly

Present (deleting a payment
may remove project information)

Eliminated (each entity can be
inserted separately)

Delete

anomaly

Eliminated (deleting payment
does not remove project/client
details)

Data

consistency

High (client, freelancer and

Low (duplicate data may
become inconsistent)
Very Low (data stored only
once)

High (single source of truth for
each entity)

Query

complexity

Low (single table, no joins) Moderate (JOINS
are required to retrieve complete information)